Talha Saleem

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Education

Sep-2022 – Jul-2026 **Quaid-i-Azam University**, Islamabad, Pakistan.
- B.S. in Bioinformatics
- CGPA: 3.0/4.0

Oct-2021 **Govt. Rizvia Islamia Degree College**, Haroonabad, Pakistan.
Oct-2021
- Pre-Medical (F.Sc.)
- Marks: 92% (& 95% in Matric Exam)

Summary

Bioinformatics graduate and AI Engineer experienced in computer vision, deep learning, and applied machine learning. Built object detection, medical imaging, synthetic data, and semiconductor defect classification pipelines using CNNs, GNNs, Qwen, and Llama. Seeking Master's research opportunities to systematically investigate machine learning problems.

Technical Skills

LLMs & VLMs Qwen3-1.7B, Llama 3.2-1B, BioBERT, Hugging Face Transformers, Ollama, NER, TF-IDF/BM25, Semantic Search, Text Classification, spaCy, Vector Search

Speech & Retrieval Faster-whisper, SpeechBrain, Speaker Diarization, CTranslate2, FAISS, Sentence Transformers

Bioinformatics Biopython, Sequence Alignment, Genomic Data Analysis, NCBI/PubMed Databases

Web Front-Backend FastAPI, WebSockets, SQLAlchemy, PostgreSQL, PyJWT, React, TypeScript, Vite, axios.

ML/MLOps & AIOps PyTorch, scikit-learn, NumPy, pandas, MLflow, DeepSpeed, Docker, Kubernetes, Nginx, Git, GitHub Actions, CI/CD, Prometheus, Grafana.

Lang & Editors Python, R, JavaScript, TypeScript, Cursor, VS Code, Jupyter Notebook

Work Experience

Jan 2026 – Present **AI Engineer**, *Upendi.app*, South Korea, Remote.
Tools: Ollama, faster-whisper, spaCy, BioBERT, Sentence Transformers, FAISS
- Trained and deployed open-weight LLMs (Qwen3-1.7B, Llama 3.2-1B) via Ollama for NLP tasks.
- Built speech-to-text and diarization pipelines using faster-whisper and SpeechBrain.
- Fine-tuned BioBERT, built ranking pipelines (spaCy, BM25) and FAISS retrieval, F1-evaluated.

Sep 2026 – Present **Full Stack AI Engineer**, *AICyro*, Islamabad, Pakistan, Onsite.
Tools: Python, PyTorch, TensorFlow, OpenAI Vision API
- Worked on the OpenAI Vision API to implement real-time image detection in the chatbot.
- Engineered a multimodal visual triage pipeline and fine-tuned their chatbot to avoid hallucinating.
- Researched network fingerprinting and built backend pipelines for VPN detection.

Sep 2025 – Dec 2025 **Associate Software Engineer**, Zeppelin Labs (SMC-Pvt), Islamabad, Pakistan.
Tools: FastAPI, SQLAlchemy, Docker, React, TypeScript, WebSockets
- Developed REST/WebSocket APIs with JWT auth, refresh tokens, and rate-limited endpoints.
- Modeled databases with SQLAlchemy ORM and containerized services using Docker and Nginx.
- Built React/TypeScript frontends with an AudioWorklet module, validated via Vitest and pytest.

Notable Projects

2026 **SemiEmapAI: Semiconductor Wafer E-Map Defect and Sensor Analytics Platform**.
Tools: PyTorch, PyTorch Geometric, scikit-learn, OpenCV, SHAP, SMOTE-ENN, Docker
- Built a BatteryML-style modular pipeline for WM-811K wafer E-maps and SECOM sensor logs.
- Trained a Multi-Head CNN (ResMHCNN) and Spatial GNN to classify 9 wafer defect patterns.
- Applied OOD confidence filtering to separate pseudo-defects from true packaging defects.
- Resolved sensor missingness and class imbalance via VAE imputation, SHAP-RFE, and SMOTE-ENN.

2026 **ResumeRanker: AI Resume Parser and Candidate Matching Platform**.
Tools: Python, spaCy, TF-IDF/BM25, Docker, PostgreSQL
- Built an NER pipeline (spaCy) to parse PDF/DOCX resumes and extract skills, education, experience.
- Designed a TF-IDF/BM25 algorithm for candidate-job relevance matching and scoring.
- Auto-tagged candidates across AI/ML, data science, and full-stack profiles.
- Deployed a Dockerized FastAPI backend, storing ranking scores and match logs in PostgreSQL.

2026 **BioSearchAI: Biomedical RAG and Semantic Literature Search Platform**.
Tools: BioBERT, Sentence Transformers, FAISS, PubMed API
- Built a document pipeline for PubMed abstracts/PDFs (extraction, chunking, cleaning).
- Trained BioBERT for biomedical entity extraction across genes, diseases, drugs, and proteins.
- Extended entity extraction further to cover organisms and clinical terminology.
- Integrated Sentence Transformers with FAISS for semantic retrieval of sentence-level evidence.

Recent Projects

Present **Vehicle Detection using custom trained YOLOv8 Model (ML/computer vision)**.
Tools: YOLOv8, TensorRT, Ultralytics, OpenCV, Pytorch, Image processing, Google Colab
- Trained a YOLOv8n model (FP16) with a synthetic, auto-labeled YOLO-format dataset pipeline.
- Benchmarked detection performance (mAP50, precision/recall) on a held-out test set.

Present **Malaria Parasite Detection System (ML/computer vision)**.
Tools: YOLOv11, SAHI, Docker, FastAPI, Streamlit
- Built a Poisson-blended synthetic pipeline and fine-tuned YOLOv11 for parasite localization.
- Integrated SAHI for tiled fast inference and deployed via Dockerized FastAPI and Streamlit.

2026 **BriefAI: Real-Time Meeting Transcription and Multilingual Summarization Platform**.
Tools: FastAPI, WebSockets, faster-whisper, Ollama (Qwen3-1.7B, Llama 3.2-1B), SpeechBrain, React, JWT
- Built a full-stack app (React, FastAPI) integrating AI services for LLM fine-tuning and inference.
- Built a live streaming pipeline via FastAPI WebSockets and browser AudioWorklet capture.
- Transcribed audio via faster-whisper and diarized speakers using SpeechBrain embeddings.
- Ran LLMs (Qwen3-1.7B and Llama 3.2-1B) via Ollama for summarization and translation tasks.

2024 **Differential Gene Expression Analysis of SCFA Treatment**.
Tools: GEOparse, scikit-learn, SciPy, Seaborn, Pandas, Matlib
- Built a GEOparse pipeline with parametric/non-parametric testing and Cohen's D correction.
- Clustered expression patterns using PCA and K-Means, visualized via volcano plots.

2023 **Protein 3D Structure Visualization**.
Tools: Biopython, RCSB PDB API, py3Dmol, Django
- Built a tool to fetch/parse PDB structures via Biopython, calculating DSSP-based features.
- Rendered interactive 3D molecular visualizations using py3Dmol.

Honors and Achievements

2025-present **Campus President**, AICP, Artificial Intelligence Community of Pakistan.
2025 **Co-Lead**, MSLA, Microsoft Student Learning Ambassadors.
2024 **Campus Ambassador**, Notion.
2023 **Campus Ambassador**, GDSC, Google Developer Student Clubs.
2021 **PEEF Scholarship**, HSSC, Punjab Educational Endowment Fund.
2019 **PEEF Scholarship**, SSC, Punjab Educational Endowment Fund.

Language Proficiency and Relevant Courseworks

Languages Fluent in English (Professional Working Proficiency) | Urdu (Native)
Coursework Object-Oriented Programming, Data Structures & Algorithms, DBMS
Coursework Machine Learning, Bioinformatics Computing, Systems Biology, Computational Genomics

Collaborators & Mentors

Dr. Adnan Ahmad Ansari
Assistant Professor,
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Dr. Junaid Rashid
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